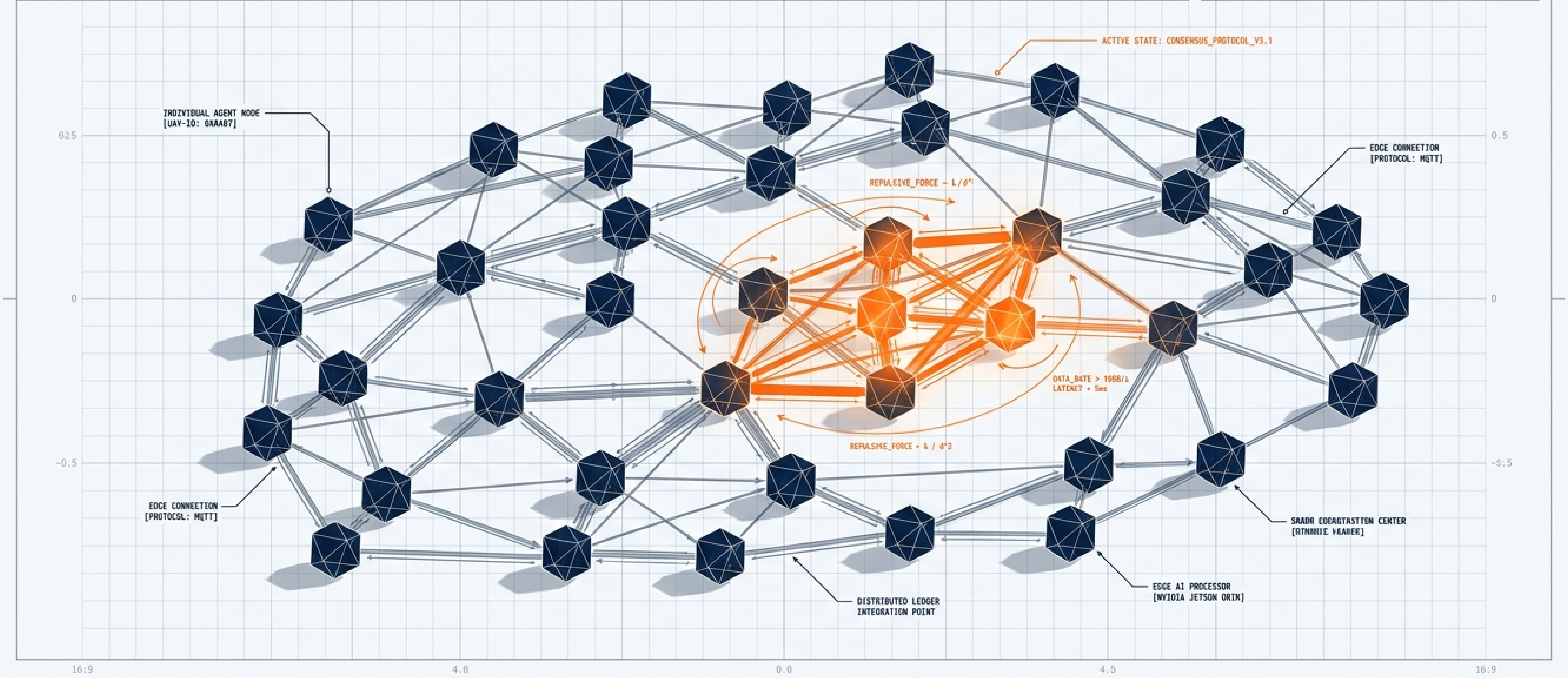
3.6 Swarm Intelligence & Coordination

Decentralized Architectures for Autonomous Multi-Agent UAV Systems

MODULE
Autonomous Systems Masterclass

FOCUS
Graph Theory, Distributed Middleware, Edge AI Integration

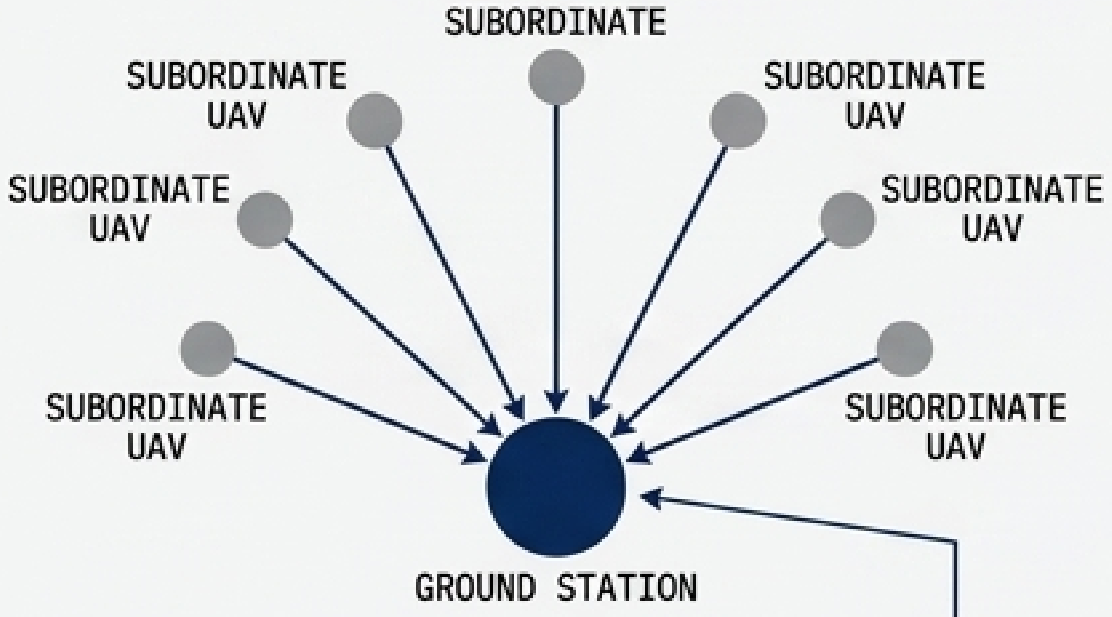


The Paradigm Shift: Architectures of Control

COMPARATIVE MATRIX: CENTRALIZED, DECENTRALIZED, AND DISTRIBUTED NETWORK TOPOLOGIES

CENTRALIZED ARCHITECTURE

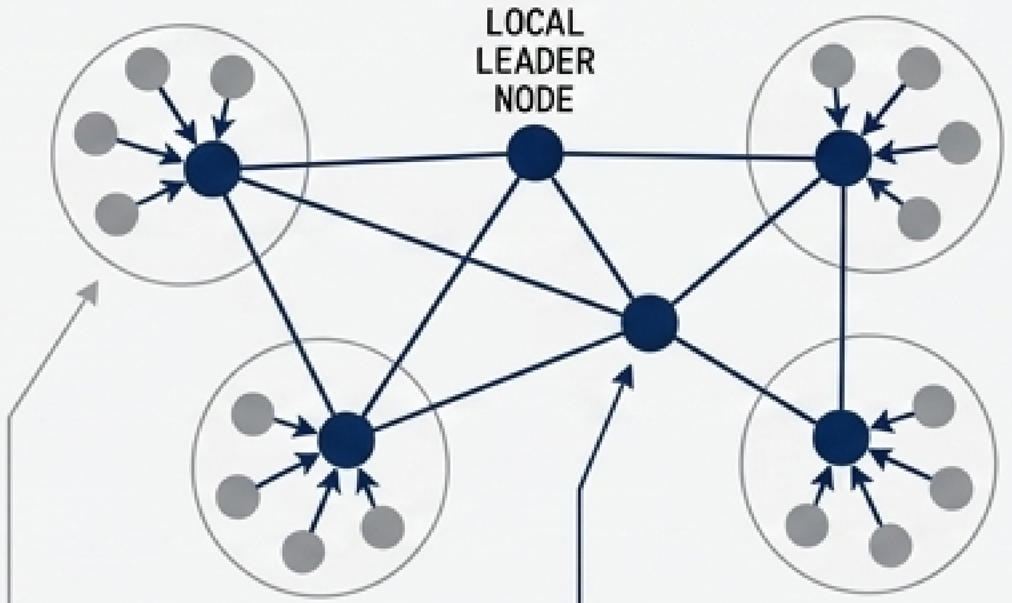
Star network topology



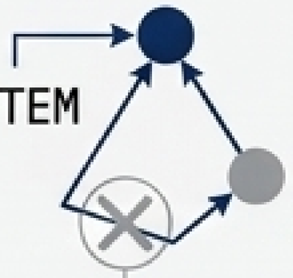
- **MECHANISM:** Single ground station directs subordinate UAVs.
- **FAILURE RISK:** CRITICAL (SINGLE POINT OF FAILURE).
- **SCALABILITY:** LOW (BANDWIDTH BOTTLENECKS).
- **ROLE:** Static, pre-programmed mission paths (e.g., Drone Light Shows).

DECENTRALIZED ARCHITECTURE

Leader-follower mesh topology



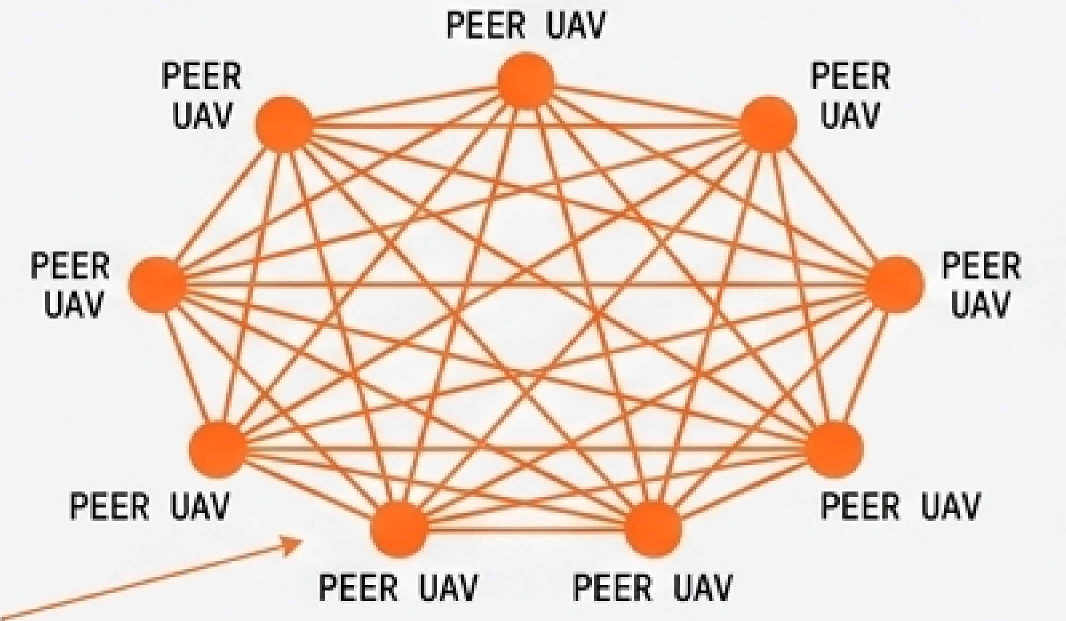
- **MECHANISM:** Clusters guided by local leader nodes acting as coordinators.
- **FAILURE RISK:** MODERATE (SYSTEM REROUTES IF LEADER FALLS).



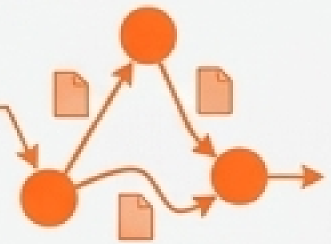
- **SCALABILITY:** MEDIUM-HIGH.
- **ROLE:** Semi-distributed tasks, agricultural monitoring.

DISTRIBUTED ARCHITECTURE

Fully connected, peer-to-peer mesh topology

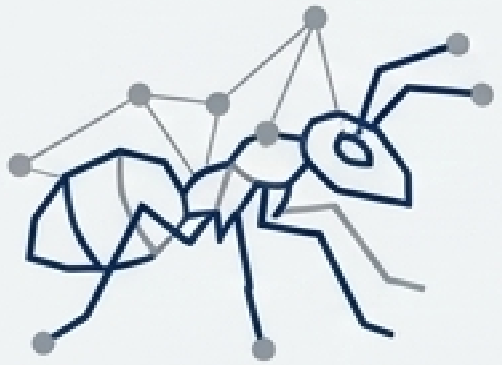


- **MECHANISM:** No global leadership. Decisions emerge from peer-to-peer data exchange and local logic.
- **FAILURE RISK:** NEAR-ZERO (SELF-HEALING).



- **SCALABILITY:** INFINITE (COMPUTE LOAD DISTRIBUTED).
- **ROLE:** Swarm Intelligence, complex dynamic environments.

Algorithmic Translation of Biological Collectives



Ant Colony Optimization (ACO)

Biological Anchor: Pheromone-based foraging and indirect communication (stigmergy).

Algorithmic Translation: Agents deposit virtual pheromones on nodes in a decision space. Subsequent agents probabilistically choose paths with higher concentrations.

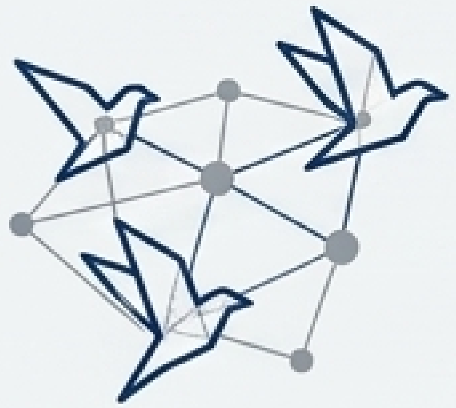
UAV Use Case:
Optimal routing, collision avoidance, and multi-objective path planning in dynamic environments.

Artificial Bee Colony (ABC)

Biological Anchor: Honey foraging and waggle-dance communication.

Algorithmic Translation: Agents are divided into employed, onlooker, and scout bees, exploring the solution space and sharing fitness information to exploit optimal targets.

UAV Use Case:
Area coverage mapping and dynamic target allocation.



Particle Swarm Optimization (PSO)

Biological Anchor: Flocking and schooling behavior.

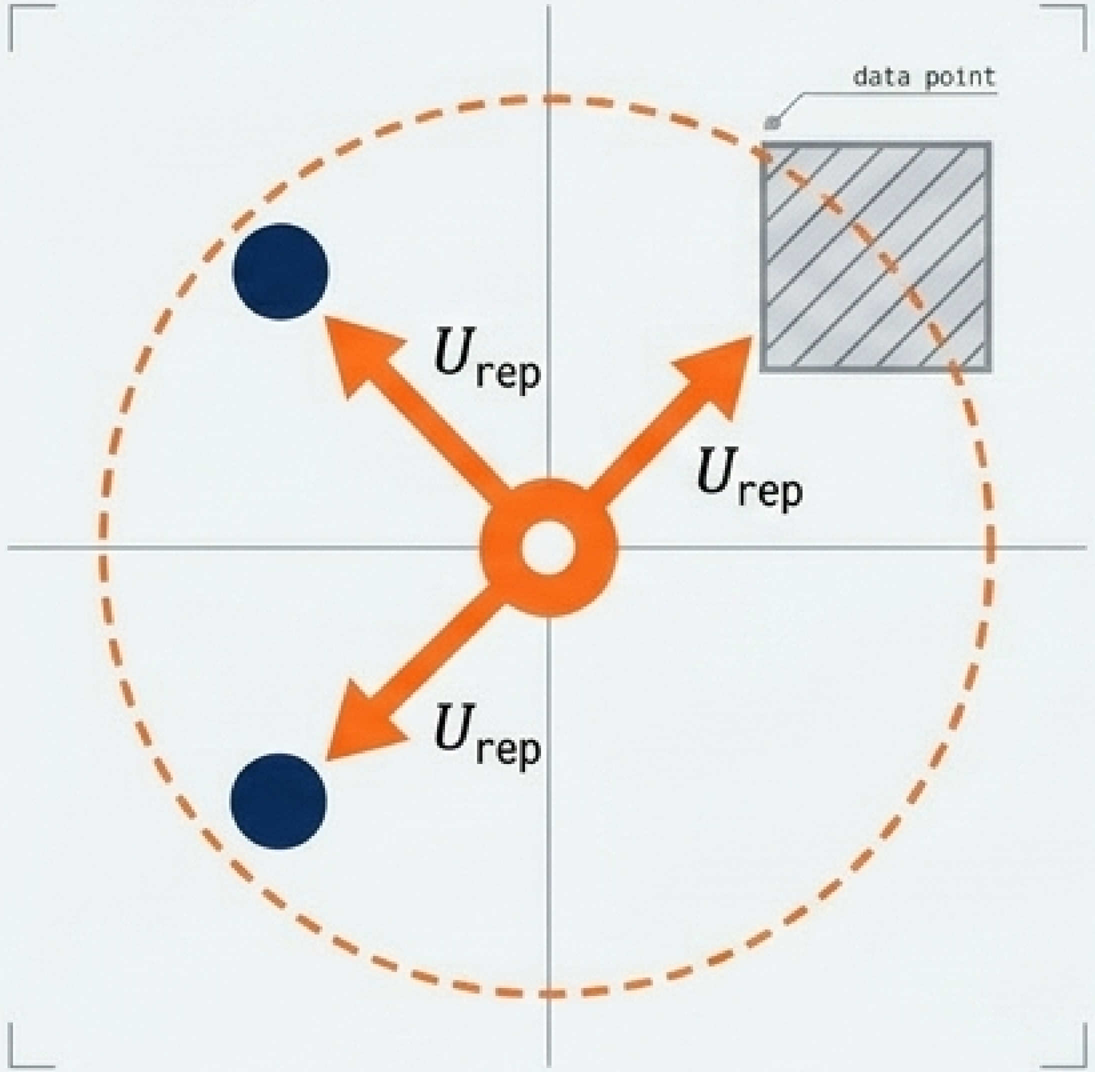
Algorithmic Translation: Simple, non-sophisticated agents adjust their trajectories based on personal best positions and the global best-known position of the swarm.

UAV Use Case:
Continuous spatial optimization, formation geometry maintenance, and cooperative localization.

Kinematic Coordination: The Boids Model & Potential Fields

TECHNICAL SCHEMATIC: BIO-INSPIRED SWARM BEHAVIORS AND POTENTIAL FIELD NAVIGATION

A Separation (Repulsive Potential)

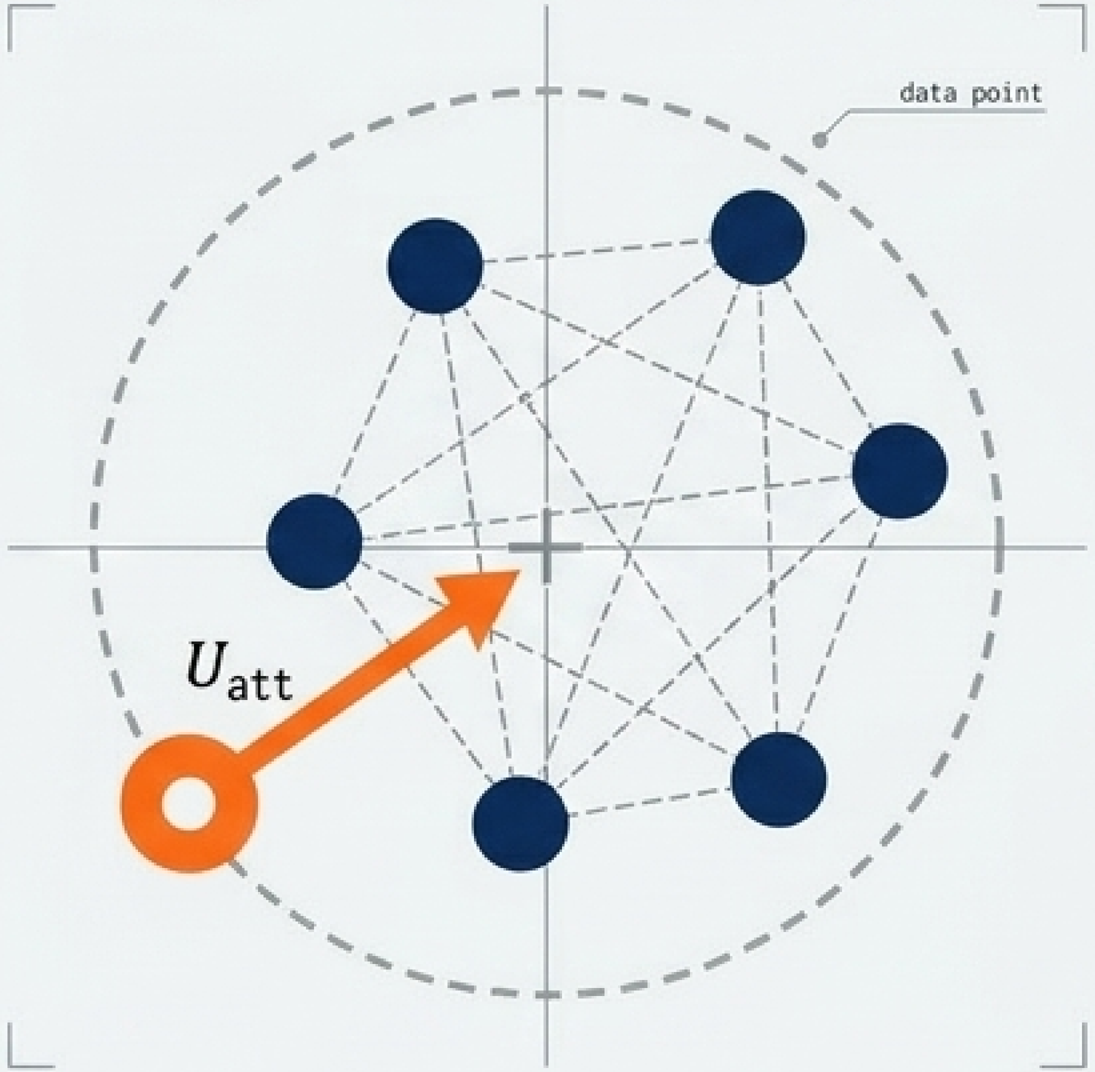


Technical Principle: Prevents intra-swarm collisions. Drones move according to artificial repulsive potential fields U_{rep} away from obstacles and peers.

B Alignment (Velocity Consensus)

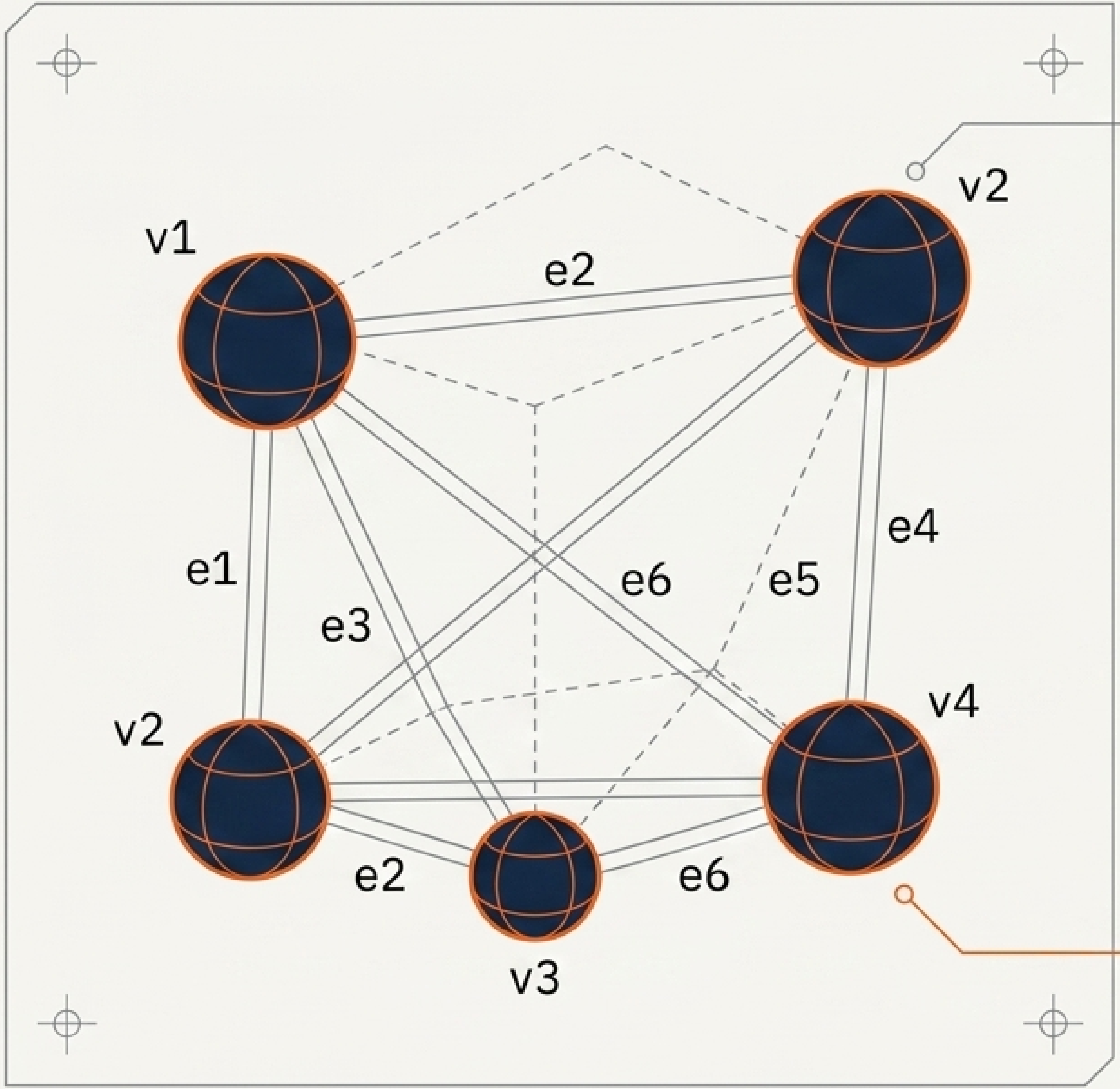
Technical Principle: Ensures directional coherence. Agents steer toward the average heading of local flockmates.

C Cohesion (Attractive Potential)



Technical Principle: Maintains swarm density. Drones follow attractive potential fields U_{att} toward a global goal or structural center.

Algebraic Representation of Swarm Networks



The Graph: Modeled as an undirected graph $G = \{V, E\}$ where V are UAV nodes and E are communication edges.

Adjacency Matrix A(G): Captures connectivity. $[A(G)]_{ij} = 1$ if UAV i and j communicate, 0 otherwise.

Degree Matrix Delta(G): Diagonal matrix capturing the total connections (degree) per UAV.

The Graph Laplacian L(G): The core mathematical engine of consensus.
L(G) = Delta(G) - A(G)

Critical Insight: The spectrum of the Laplacian matrix governs swarm dynamics. The Fiedler eigenvalue (lambda_2) dictates the rate of convergence—a larger lambda_2 means the swarm reaches consensus faster.

Accelerated Consensus via Neighboring Velocity Information

$$u_i(t) = -\sum_{i\sim j}(x_i(t) - x_j(t)) + \gamma \sum_{i\sim j} \dot{x}_j(t),$$

Control input for agent i.

Standard relative position error.

The breakthrough variable: Neighboring Velocity integration.

The Problem:

High control gains increase convergence rate but also increase the maximum singular value and eigenvalue of the system, violating **control saturation limits**.

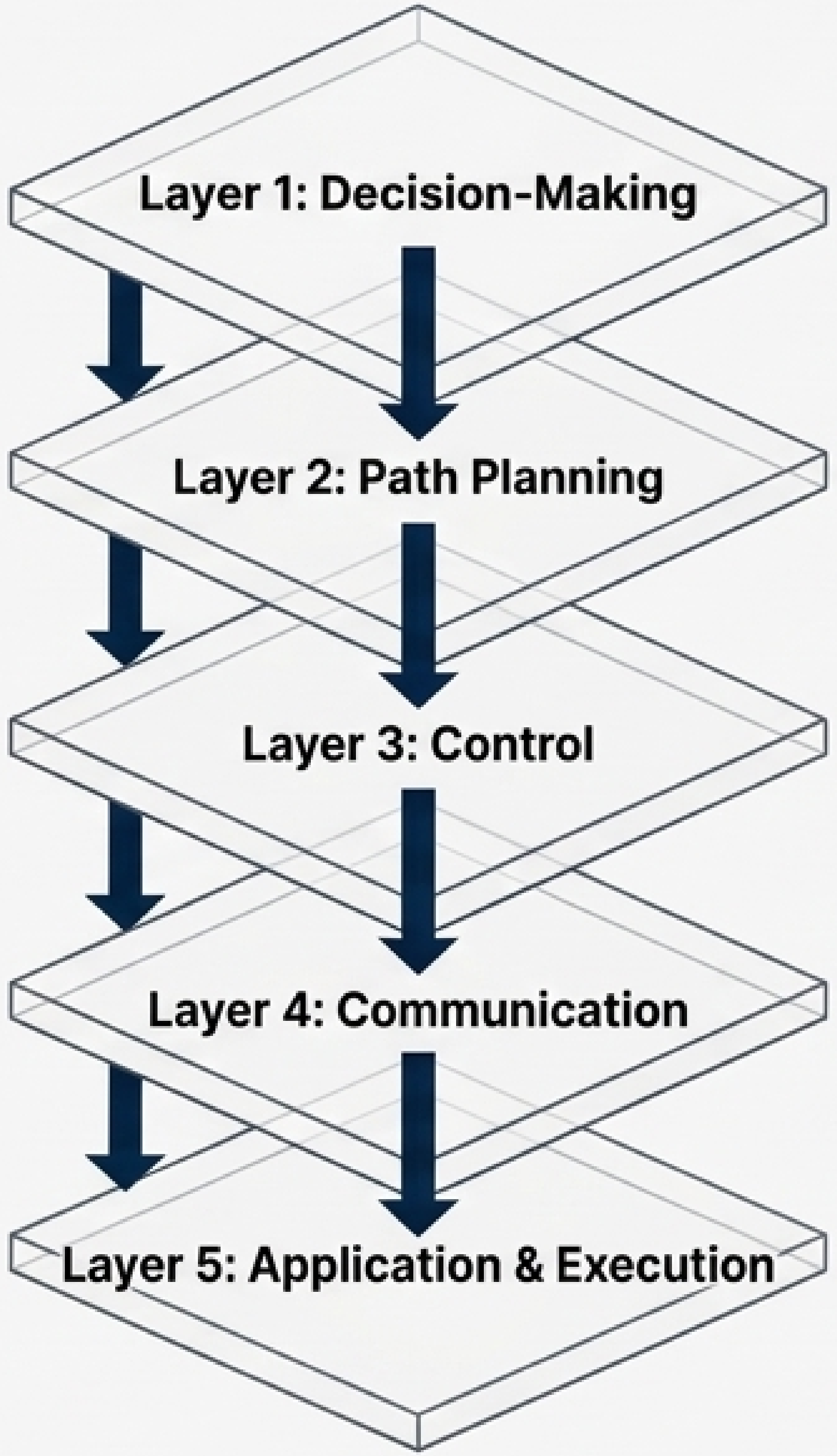
The Solution:

Adding neighboring velocity $\dot{x}_j(t)$ allows an increase in convergence rate without increasing the maximum eigenvalue of the graph Laplacian.

Result:

A connected, undirected graph topology mimics the **efficiency of a weighted complete graph topology**, bypassing traditional Fiedler bounds.

The 5-Layer Swarm Engineering Architecture



Layer 1: Decision-Making
Role: Strategic core, mission planning, task decomposition.
Tech: Cognitive AI, Multi-Agent Reinforcement Learning (MARL).

Layer 2: Path Planning
Role: Spatiotemporal trajectory synchronization.
Tech: Dubins curves (for kinematic constraints), Distributed Model Predictive Control (DMPC).

Layer 3: Control
Role: Low-level execution, safe inter-drone distances.
Tech: Real-time PID control loops (1-2 kHz), Potential Field algorithms.

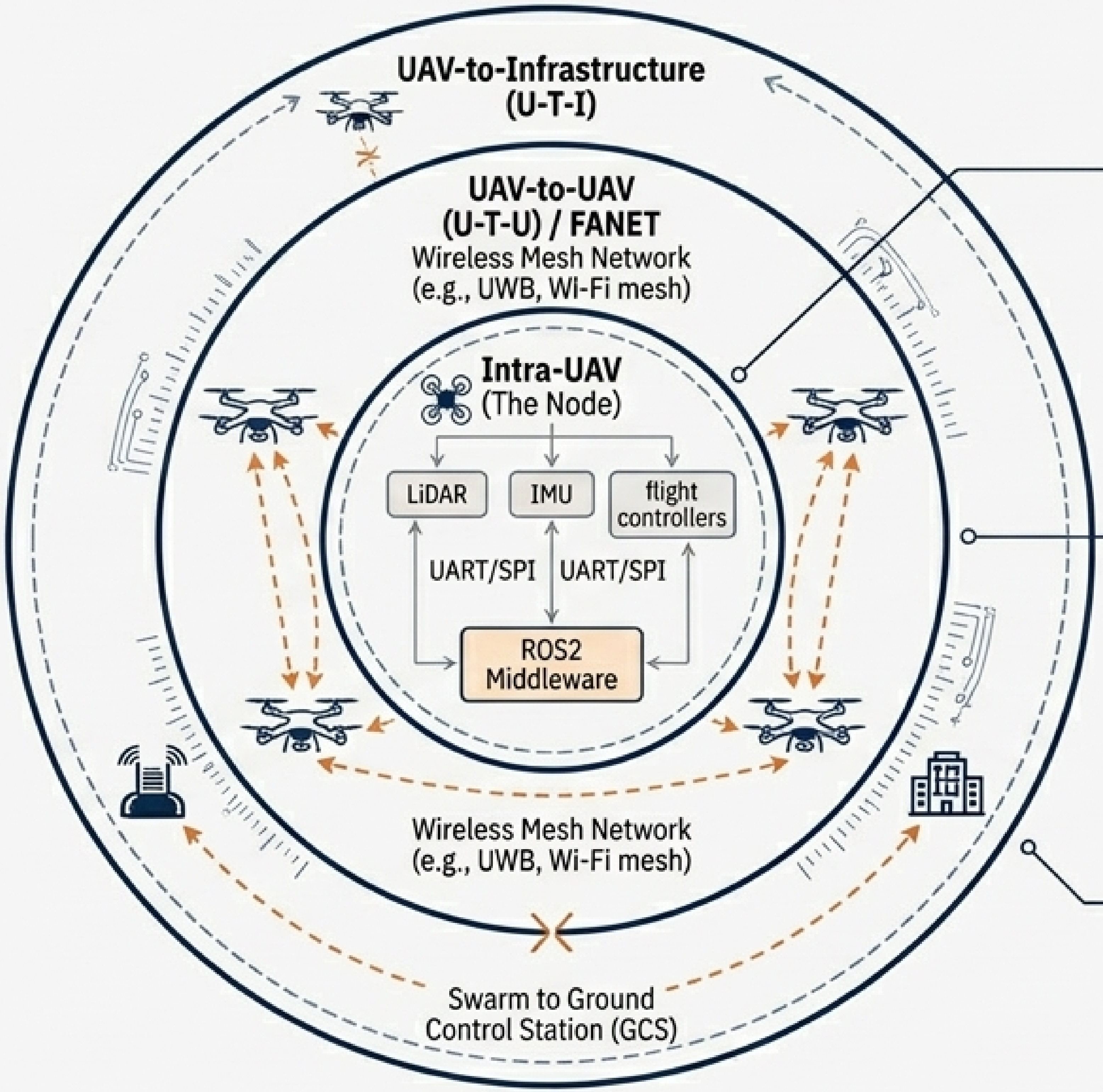
Layer 4: Communication
Role: Flying Ad-hoc Networks (FANET) infrastructure.
Tech: Mesh routing, topology stabilization in high-mobility nodes.

Layer 5: Application & Execution
Role: Environmental interaction.
Tech: Subsystem actuators, LIDAR, multispectral payloads.

Middleware Protocol Diagnostic Matrix

	MAVLink	DDS (Data Distribution Service)	MQTT / ZeroMQ
Architecture Type	Binary-serialized, publish-subscribe	Data-centric pub/sub (Fast DDS, Cyclone)	Message queuing
Overhead	Ultra-low (header-only)	Medium	Low
Real-Time Determinism	Low (Basic telemetry)	High (QoS parameters ensure soft/hard real-time)	Non-deterministic
Scalability	Poor for dense data (bottlenecks)	Extreme (Highly distributed partitioned systems)	High
Best Use Case	Micro-air vehicles, low-resource flight controllers	Complex multi-agent swarms, ROS2 integration	Lightweight IPC, non-critical telemetry to ground stations

Data Flow & FANET Topologies



Ring 1: Intra-UAV (The Node)

Scope: Internal bus communication.

Function: UART/SPI connections linking LiDAR, IMU, and flight controllers to the companion computer via ROS2 middleware.

Ring 2: UAV-to-UAV (U-T-U) / FANET

Scope: Wireless Mesh Network (e.g., UWB, Wi-Fi mesh).

Function: Decentralized peer-to-peer exchange of kinematic vectors, collision alerts, and consensus states. Self-healing topology prevents single-point failure.

Ring 3: UAV-to-Infrastructure (U-T-I)

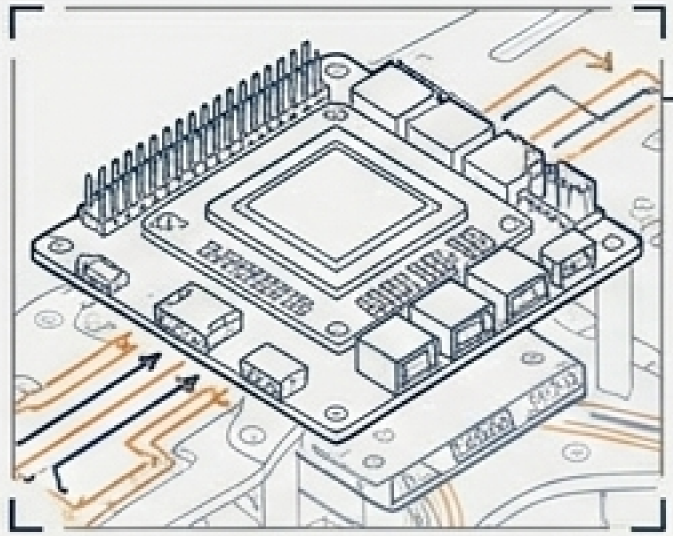
Scope: Swarm to Ground Control Station (GCS).

Function: Low-bandwidth mission updates, asynchronous telemetry logging. Designed to be safely severed in contested/denied environments.

Hardware-Software Co-Design: The UAV Node

High-Level Autonomy

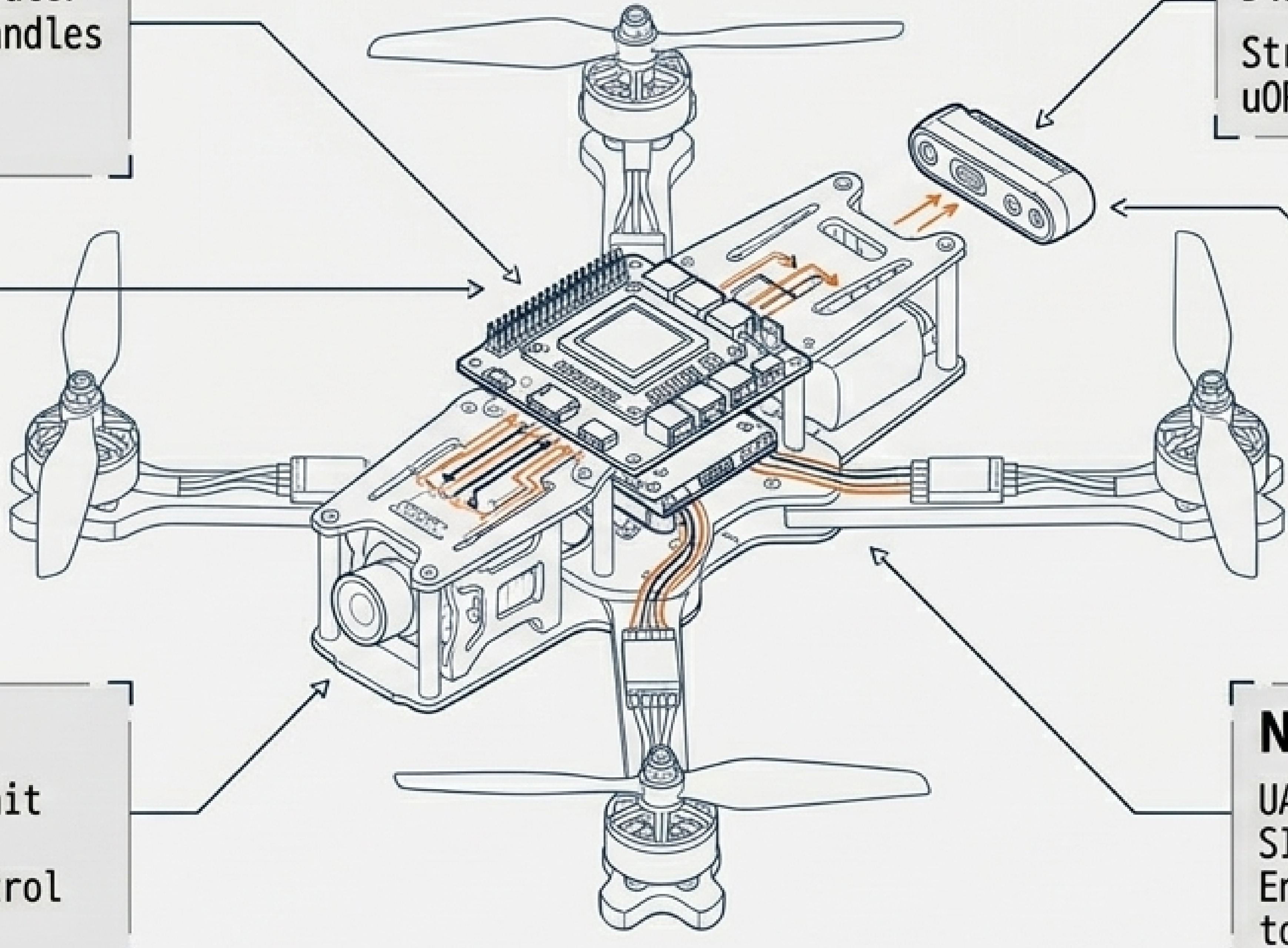
NVIDIA Jetson Companion Computer running Linux PREEMPT_RT. Handles SLAM, perception, and DMPC optimization at 1-10 Hz.

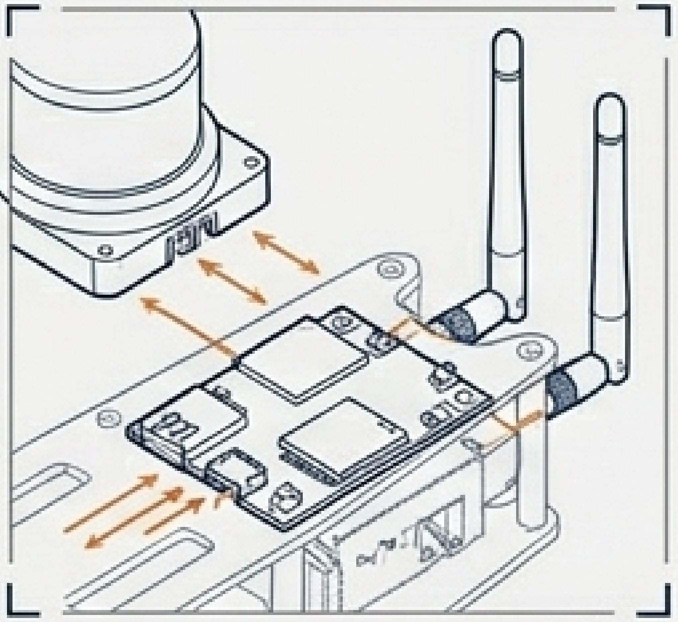


Perception Layer

D435i RealSense Camera / LIDAR.

Streams point clouds via uORB/ROS2 topics.





Low-Level Execution

Pixhawk 6C Flight Control Unit (FCU). Runs RTOS/firmware. Handles real-time motor control loops at 1-2 kHz.

Networking Hardware

UAVConnector / AX210 / SIM8200EA.
Enables dynamic U-T-U mesh topology.

Synthesis: The Agentic Swarm Loop

Act (Control Layer)

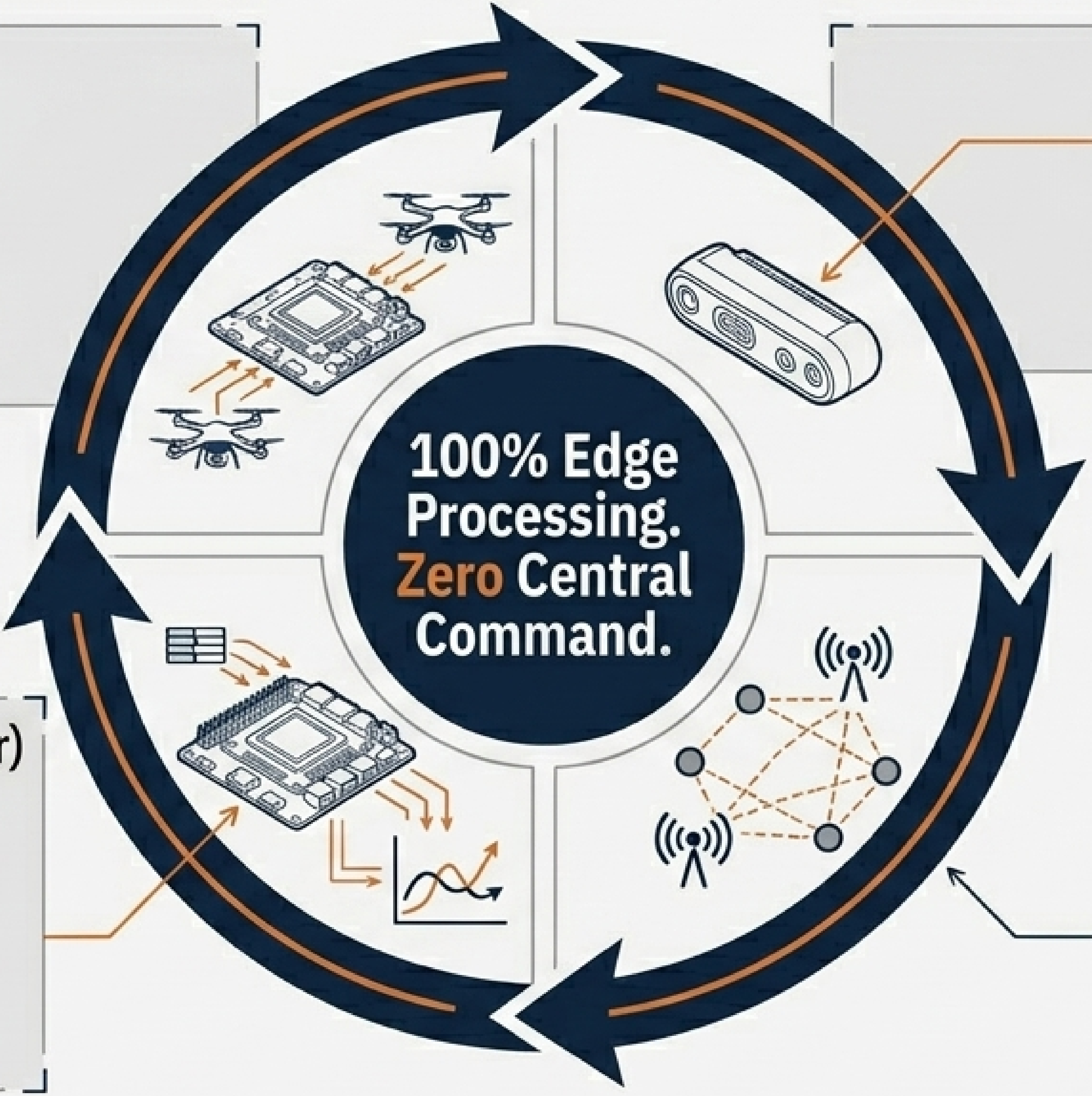
Translated motor commands sent to Pixhawk FCU.

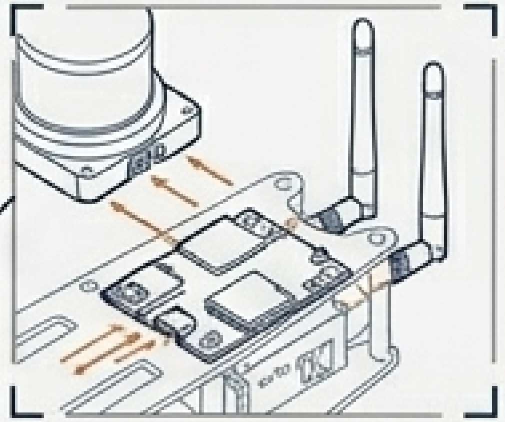
Quadcopter adjusts pitch/roll/yaw to maintain swarm geometry.

Sense (Perception Layer)

D435i Camera/LIDAR acquires environmental data.

Generates point clouds and obstacle state vectors.





Compute (Decision/Path Layer)

NVIDIA Jetson processes incoming peer data.

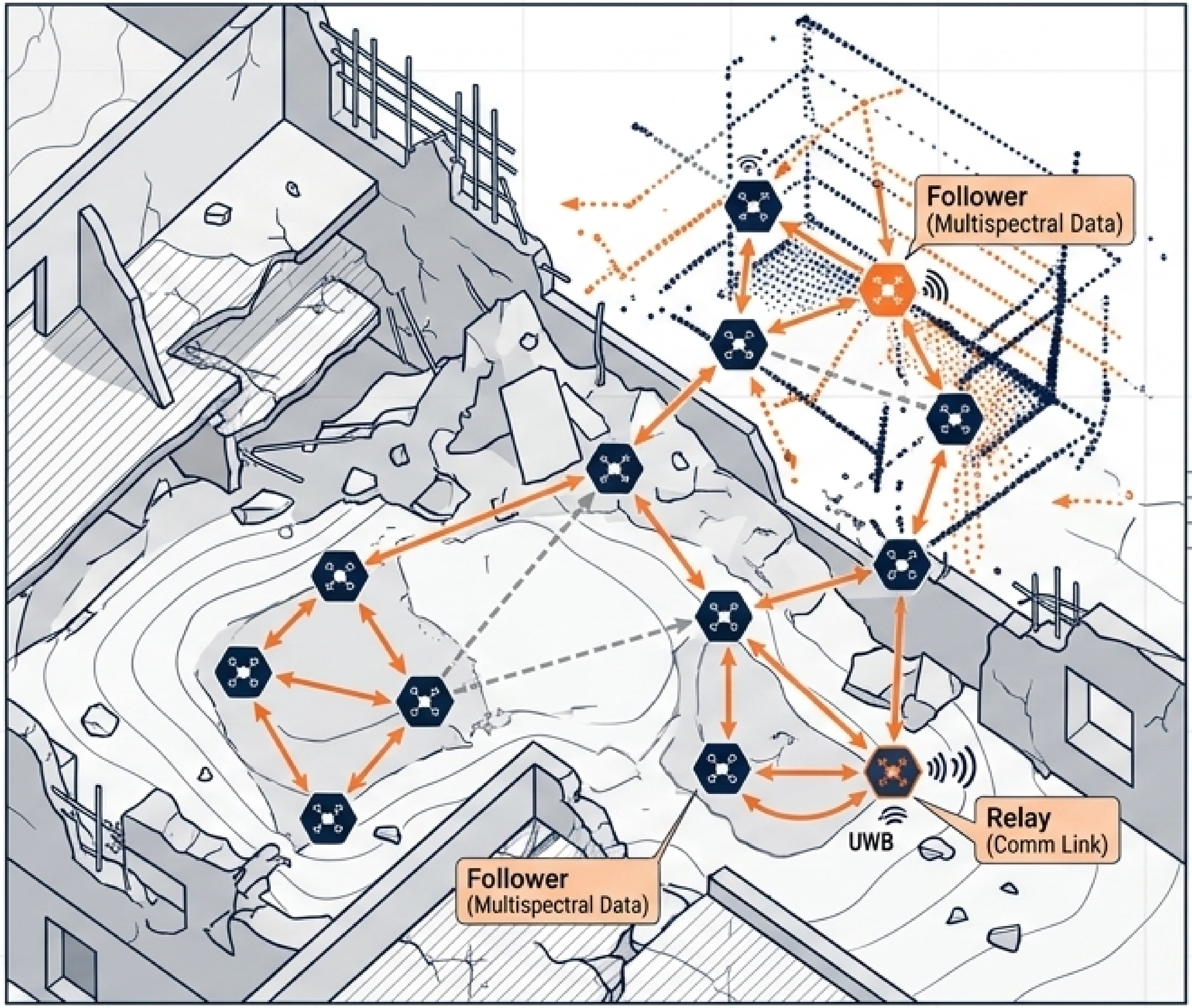
Graph Laplacian $L(G)$ is solved locally to update the Boids cohesive/repulsive

Share (Comm Layer)

State vectors broadcast via Fast DDS over FANET to neighboring nodes.

Field Application: GNSS-Denied Target Allocation

Precision Engineering & Technical Schematic



The Scenario

The Mission

Post-disaster rapid mapping in a GPS-denied, **communication-contested** environment.

The Challenge

Traditional full-connected networks waste bandwidth and fail without line-of-sight.

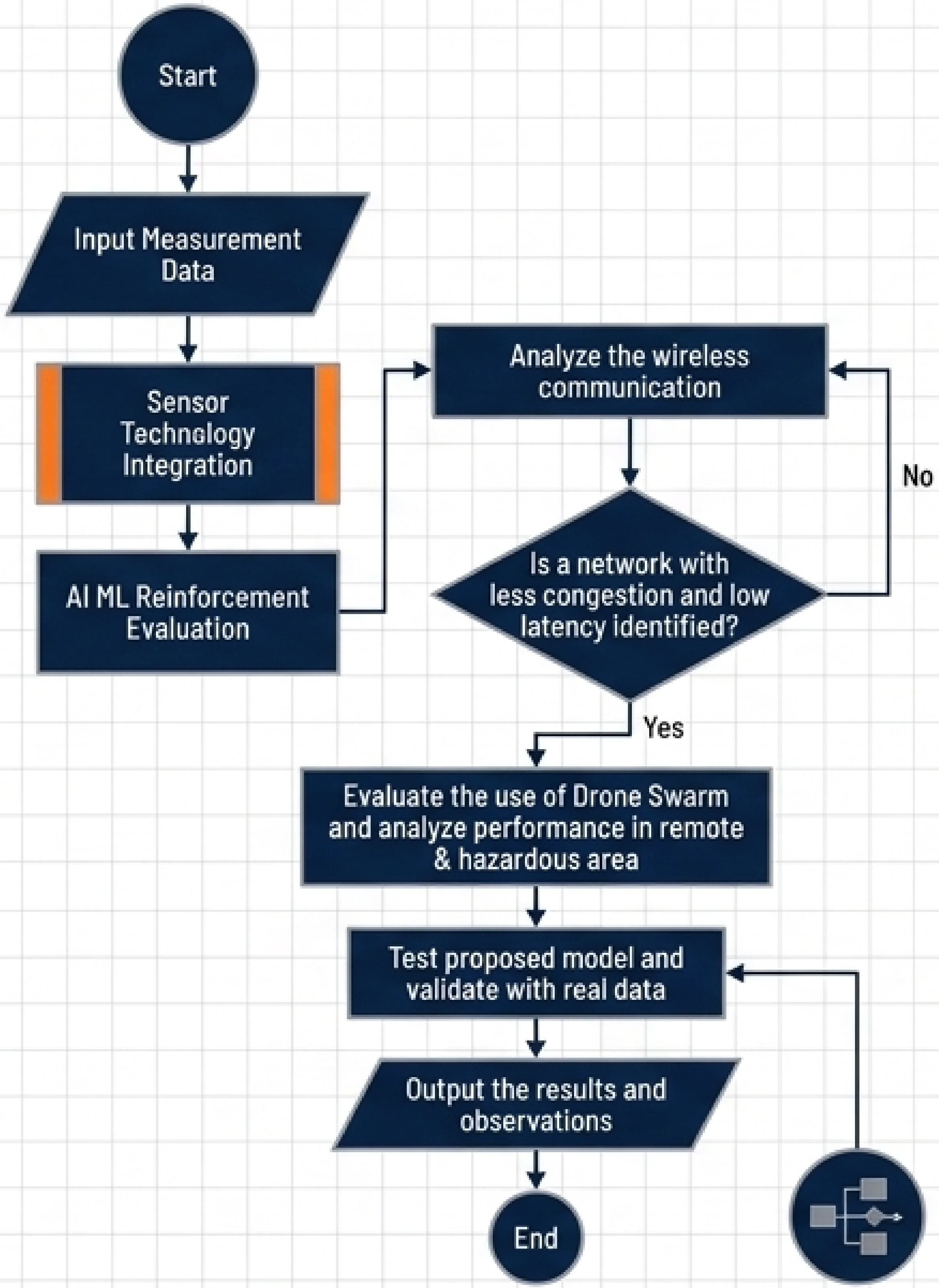
The Solution

Game-Connected Cooperative Strategy. Drones autonomously form task graphs and negotiate connections based on a spatial proximity coalition formation game.

Result

Swarm dynamically allocates roles (e.g., Scout mapping ahead, Followers collecting high-res multispectral data) while maintaining inter-drone relative localization via UWB and visual-inertial state estimation.

Horizon 1: Multi-Agent Reinforcement Learning



Beyond Heuristics

Standard ACO/PSO algorithms struggle with high-dimensional, unpredictable environments. The solution is hybrid models like RL-PSO.

Deep Deterministic Policy Gradient (DDPG)

Used to optimize conflicting objectives (task balance, flight distance, energy cost) in large, discrete action spaces.

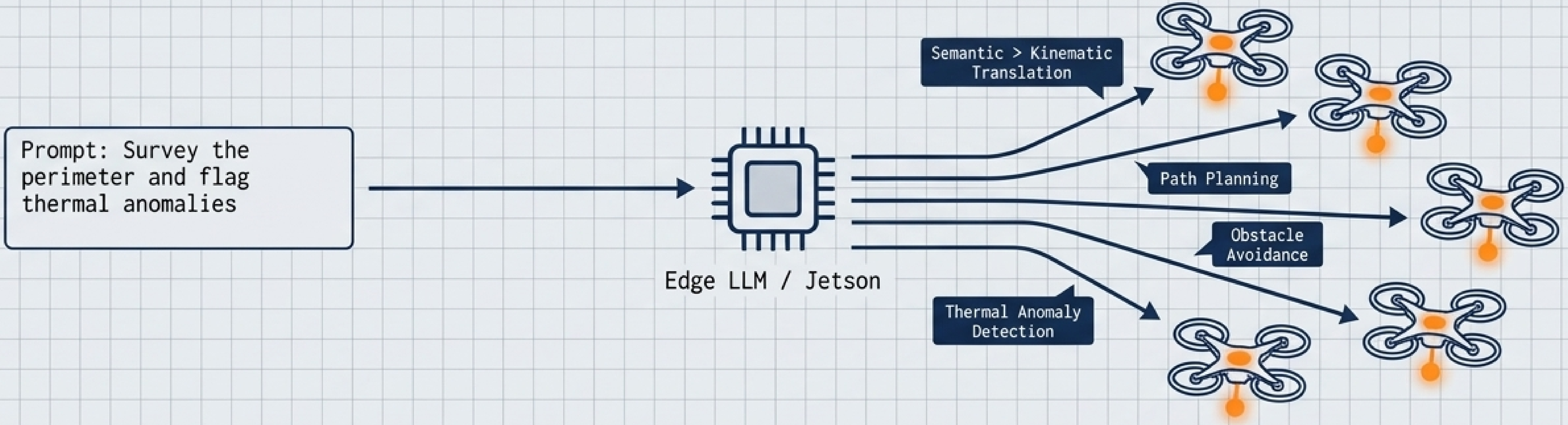
Goal Consistency

Multi-head soft attention algorithms (GCMSA) allow swarms to execute dynamic target rounding and obstacle avoidance with >82% success rates.

Energy Optimization

RL models actively learn to mitigate wind-induced perturbations and battery discharge curves, maximizing overall mission endurance.

Horizon 2: Agentic AI & Edge LLMs



The Semantic Swarm

Transitioning from mathematical waypoints to natural language mission objectives, allowing operators to command swarms via semantic reasoning.

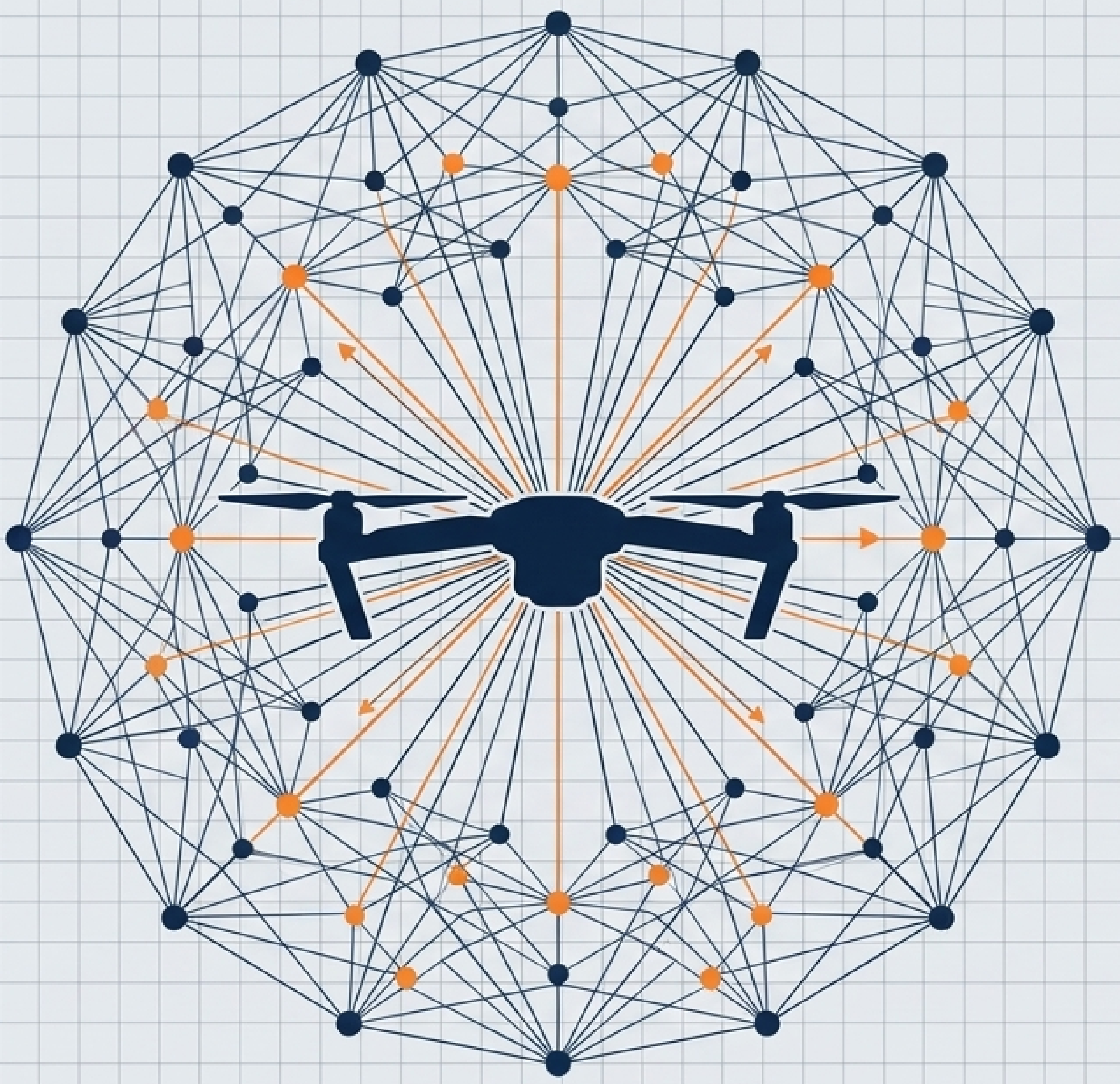
Lightweight Embedded LLMs

Running highly quantized reasoning models directly on edge hardware to interpret mission semantics without relying on cloud infrastructure or high-bandwidth links.

Heterogeneous Systems

Future FANET standards (IEEE 1937.1) enabling seamless interoperability between aerial swarms, ground vehicles (UGVs), and surface vehicles (USVs) in a unified intelligence network.

The Future of Aerial Architecture is Decentralized



1. Algorithmic Superiority

Graph theory and consensus protocols outmaneuver centralized bottlenecks in speed, scalability, and resilience.

2. Hardware Readiness

Edge AI computing and deterministic middleware (DDS`) have closed the gap between theoretical math and physical execution.

3. Emergent Capability

The true power of the swarm is not in the complexity of the drone, but in the intelligence of the network. The flock is the computer.